

SportSlotReservation

Phase 4 Engineering Report — Frontend Foundation

Field	Value
Scope	React PWA: auth, tenant resolution, booking lifecycle UI, per-tenant branding, public deploy
Period	2026-06-13 (sub-phases 4.1–4.6, incl. 4.6.1)
Author	Chandra Nakkalakunta — Principal Cloud & AI/ML Architect, Chandra AI Labs
Status	PHASE 4 COMPLETE — live at sport-slot-dev.web.app (custom domain deferred to Phase 7)
Frontend tests	15 (vitest); backend 85, coverage 90%
Stack	React 18 · Vite 6 · TypeScript · TanStack Query · Firebase Auth · vite-plugin-pwa · Firebase Hosting
ADRs	0012 Frontend Architecture · 0013 Error Presentation & i18n

1. Executive Summary

Phase 4 delivered the resident-facing Progressive Web App and put SportSlot on the public internet. A React 18 + Vite + TypeScript PWA provides Firebase authentication, hostname- and claim-based tenant resolution, the full booking lifecycle (browse facilities, computed-availability grid, book, view, cancel), and per-tenant branding driven from the tenant document. The app is served by Firebase Hosting at `sport-slot-dev.web.app`, with all API traffic proxied same-origin to Cloud Run via Hosting rewrites — eliminating CORS entirely.

Every backend guard built in Phases 2–3 was exercised through the UI and validated over the public edge: role gating, the booking window and horizon, quota, slot contention, fail-closed locking, and buffer-enforced cancellation — each surfaced to the user through a code-keyed, i18n-ready error catalog. The custom wildcard domain was deliberately deferred to Phase 7, where the load balancer and certificate it requires are already planned.

2. Key Decisions

Decision	Choice	Rationale
Hosting	Firebase Hosting; named subdomains now, LB wildcard Phase 7	Classic Hosting caps 20 subdomains/apex and has no wildcard; the LB (needed for Cloud Armor) brings the production wildcard
API routing	Same-origin via Hosting rewrites <code>/api/**</code> → Cloud Run	Zero CORS surface; original host carried as <code>X-Forwarded-Host</code>
State	TanStack Query (server) + React context (auth/tenant)	No store library until complexity earns it
Theming	CSS custom properties, applied at runtime from tenant doc	Per-tenant branding without a build-time theme; Tailwind rejected for fighting runtime switching
Error UI	Code-keyed catalog, i18n-ready; resolver anticipates locale + tenant overrides	Wording is data not code; localization is additive (ADR-0013)
Branding source	Public unauthenticated endpoint <code>/tenants/{slug}/branding</code>	Sign-in page must be branded before a token exists; cosmetic fields only
Cancellable flag	Computed server-side, returned per booking	The backend owns the buffer rule; the client never duplicates policy
Tenant cross-check	<code>X-Forwarded-Host</code> when present; host-check conditional, JWT authoritative	Bare hosting hosts trust the signed claim; spoofing cannot cross tenants (ADR-0007/0012)

3. Issue Log

#	Issue	Rule adopted
1	Phase 1 stub files ambushed a pre-flight (4th occurrence)	Pre-flights on a first-touched dir expect stubs, not absence
2	New frontend/src/lib/ silently un-committed by unanchored <code>.gitignore</code> 'lib/'	Gitignore patterns in a polyglot monorepo must be path-anchored
3	Facilities load hung then 500'd — expired local ADC	Dev ADC expires by design (no static keys); re-auth is routine, documented
4	Booking 409 produced no visible feedback	Validate the error path in the UI, not just the happy path
5	Public <code>.web.app</code> 403'd every call — deployed image predated the middleware change	Deployed frontend and backend are independently versioned; redeploy before validating
6	App stayed default-blue — branding never fetched on <code>.web.app</code>	A new surface (bare hosting host) needs the same tenant-resolution path the backend got; mirror cross-layer rules

4. Verify-First Wins

Two design-time documentation checks prevented architectural dead-ends. First, Firebase classic Hosting's 20-subdomain limit and absence of wildcard custom domains was caught before committing to it — redirecting the design to named subdomains now and a load-balancer wildcard in Phase 7. Second, the mechanism by which the original hostname survives a Hosting-to-Cloud-Run rewrite (the X-Forwarded-Host header) was confirmed against current documentation before the tenant cross-check was modified, rather than discovered during debugging. Both follow the standing rule: verify platform behavior at the installed/current version before designing on it.

5. Public-Edge Validation

Flow	Result
Sign-in (Firebase, email/password)	Token minted with Phase 2 custom claims; tenant resolved
Facility list + availability grid	Computed slots with tenant-clock states (past/booked/window/horizon/in-progress)
Booking	Created via same-origin rewrite → Cloud Run → Firestore; grid auto-refreshes
Guard messages	Quota / contention / already-booked surfaced as friendly catalog text
Cancellation	Buffer enforced server-side; non-cancellable bookings hide the action
Per-tenant branding	Tenant doc colors applied to CSS variables pre- and post-login

6. Phase 5 Preview — CI/CD

Phase 4 was deployed through manual, ordered make commands (build-push, deploy-dev, deploy-hosting, seed). Phase 5 automates this: GitHub Actions authenticating via Workload Identity Federation (no service-account keys — the charter's zero-static-credentials rule extended to CI), running the existing gate suite, building, and deploying both surfaces on push to main. This directly eliminates two failure modes felt during Phase 4 — a skipped manual step and a stale deployed image — by making 'committed' and 'deployed' the same event. Binary Authorization and vulnerability scanning round out the supply-chain hardening.